

B2B Sales Org Developer Best Practice



September 28th, 2021

Rules of Engagement

- ✓ Please participate – we'd like this to be interactive!
- ✓ Avoid the temptation to check email – focus on the training
- ✓ There are no dumb questions – chances are someone else is thinking it too!



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DEVOPS PROCESS OVERVIEW

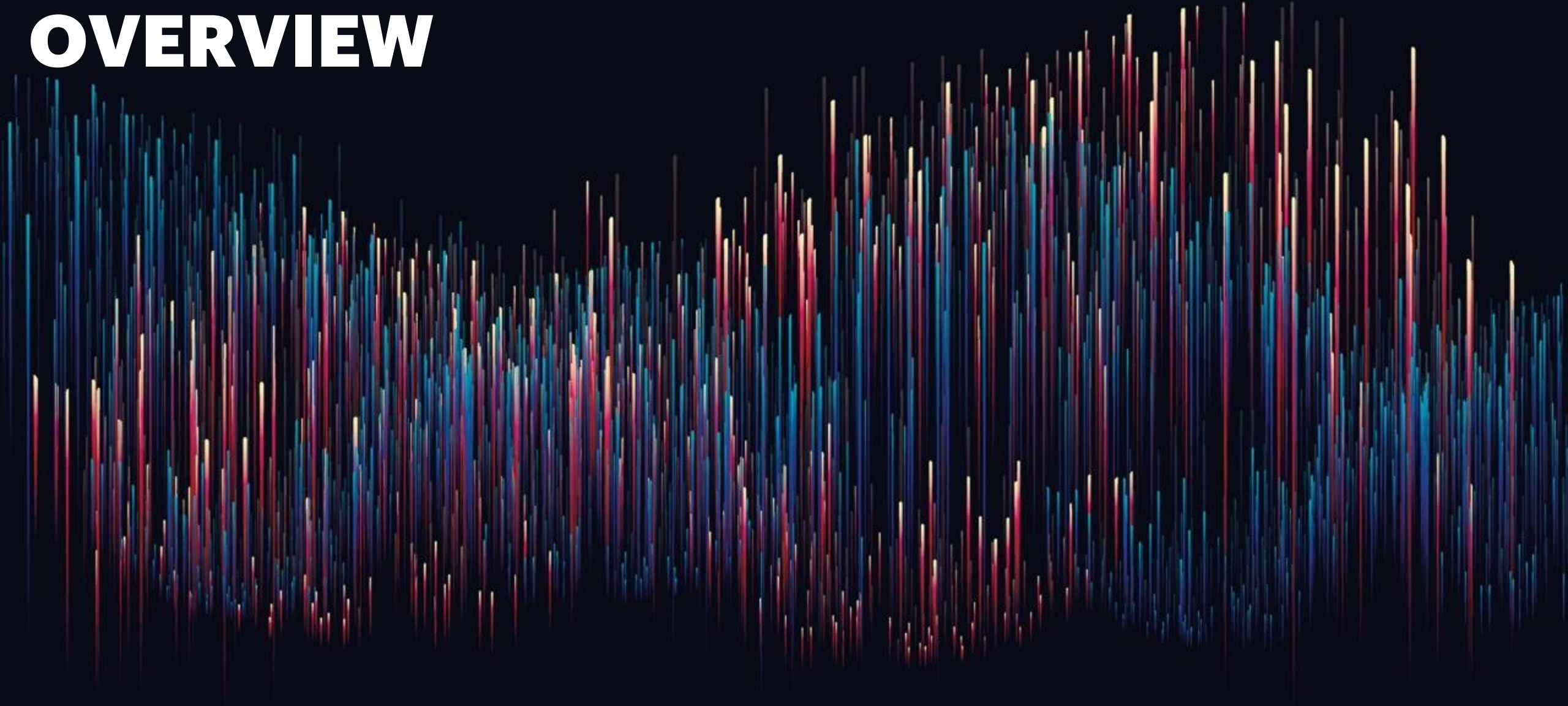


COPADO BEST PRACTICES



NEXT STEPS

DEVOPS PROCESS OVERVIEW



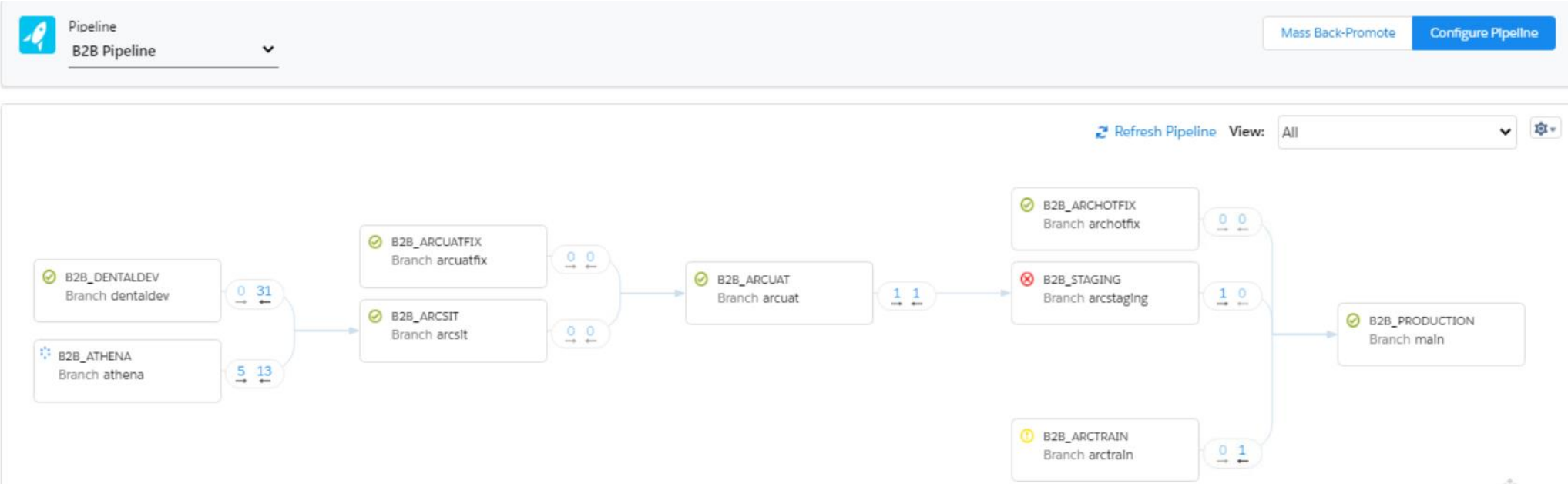
DEVOPS PROCESS OVERVIEW

DevOps Tools and Processes

- **DevOps Tools**
 - **Azure Board**
 - **Azure Repo**
 - **Copado**
- **Tools Integration:**
 - ✓ **Azure board** – Azure board for Sprint Planning, User Story and Bug Fix tracking for B2B Sales Org (Group CRM and Dental)
 - ✓ **Azure Repo** – Azure repo used for source code management
 - ✓ **Copado Pipeline** – B2B end-to-end pipeline management
 - ✓ **Azure/Copado Sync** – To sync User Stories between Azure and Copado
 - ✓ Scheduled batch job at 12AM EST
 - ✓ Manual Sync as needed

Pipeline Architecture

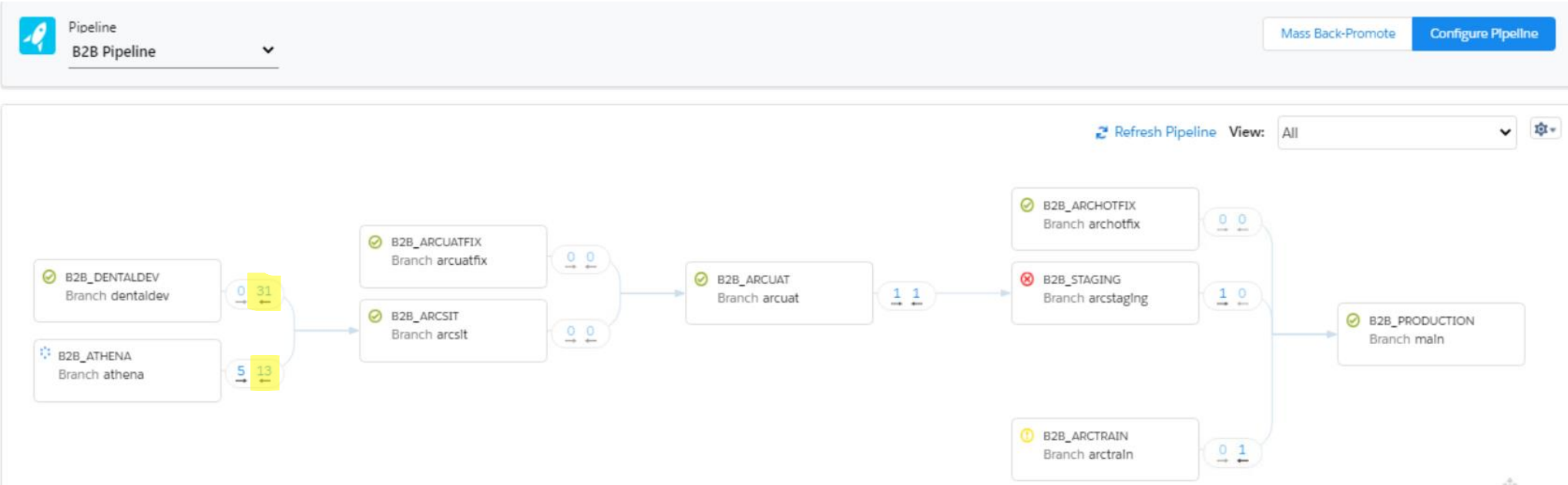
✓ Copado User Story Progression Path



Forward Promotion	Dev-to-SIT	SIT-to-UAT	UAT-to-Staging	Staging-to-Production
Responsibility	Developer	DevOps	DevOps	DevOps

Pipeline Architecture

✓ Copado User Story Back-Promotion

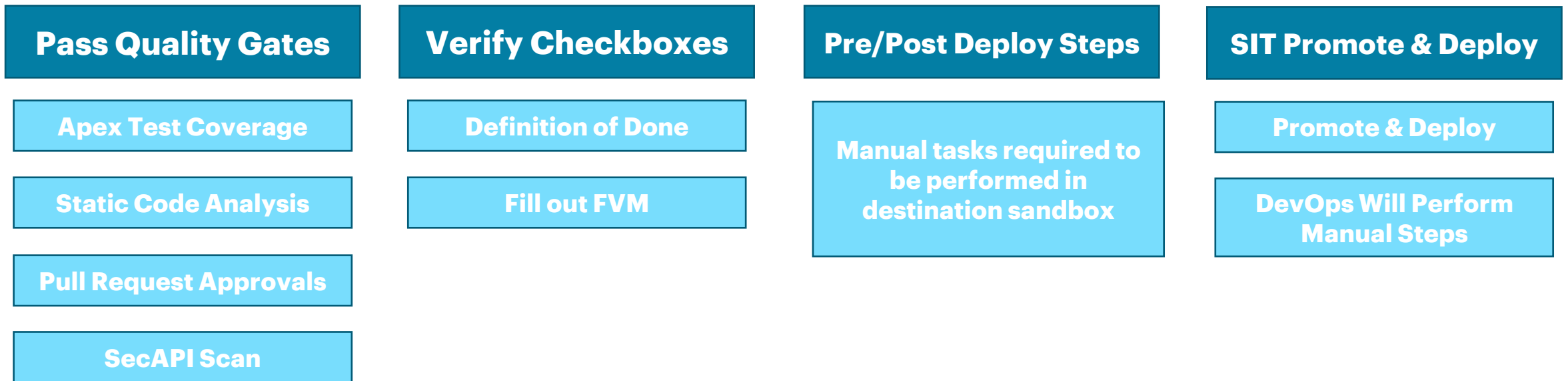


Back-Promotion	SIT-to-Dev	UAT-UATFix	Prod-Hotfix/Arctrain
Responsibility	Developer	DevOps	DevOps

User Story Deployment Readiness



User Story deployment readiness to SIT



Pre/Post-Deployment Manual Tasks

✓ **Where to put instructions**

- ❖ Document all instructions in the "Description" section of the manual task
- ❖ If the process is complex and requires screenshots, please attach as a word document with detailed instructions

✓ **Guidelines for Detail Level of Instructions**

- ❖ Ensure that instructions are easy and obvious for DevOps to be able to perform without any support from Dev Teams
 - ❖ This eases the process for DevOps and ensure timely deployment windows

Quality Gates

✓ Copado User Story Level

- ❖ Apex test coverage of 85%
- ❖ SCA – Priority 1 violations will block deployments
- ❖ Pull request Approvals
 - ❖ 1 Level of Approval if story has no Security components (Approval from Dev Lead)
 - ❖ 2 Levels of Approval if story has Security components (Approval from Dev Lead & Security Team)
- ❖ SecAPI scan

✓ Copado Pipeline Level

- ❖ Dev-to-SIT
 - ❖ Stories with Apex Components to go through Run All Tests in SIT
 - ❖ Stories without Apex components run no tests
- ❖ SecAPI scan at branch level for Main and UAT (Performed by Security Team)

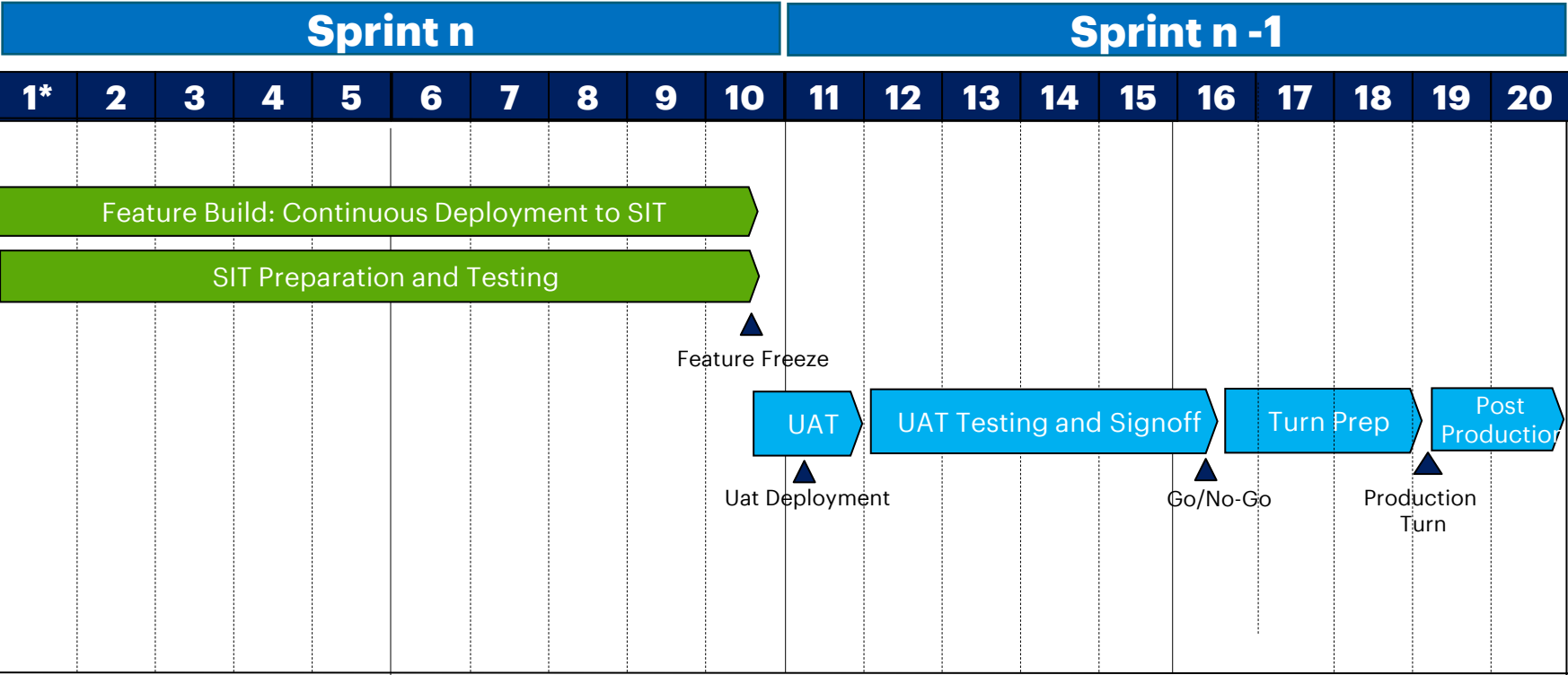
Reference Links

- ✓ **E3 site for Copado Documentation and DevOps process:** [go/learncopado](#)
- ✓ **Static Code Analysis (SCA)** : <https://devopsdocs.apps.sdc-prod-cf.humana.com/products/copado/enhancements/pmdpr/>
- ✓ **Pull Requests** : <https://devopsdocs.apps.sdc-prod-cf.humana.com/products/copado/enhancements/pr/>
- ✓ **SecAPI**: <https://devopsdocs.apps.sdc-prod-cf.humana.com/products/copado/enhancements/secapi/>
- ✓ **FVM** : <https://devopsdocs.apps.sdc-prod-cf.humana.com/products/copado/enhancements/fvm/>

DevOps Support Requests

- ✓ **These are the things DevOps help with:**
 - ✓ **Azure/Copado Sync**
 - ✓ **Manual Task Support**
 - ✓ **Deployment Issues and troubleshooting**
- ✓ **Things out of DevOps Scope:**
 - ✓ **Copado Concerns:**
 - ✓ **Create E3 Ticket at ([go/csd](#))**

Sprint Execution Timeline



**indicates business days*

Cut off Time Guide

- ✓ Stories must be in 'SIT Done' Azure Board column by **Friday, 2 PM EST on the last day of the sprint** to be picked for UAT deployment.
- ✓ Stories must be deployed to UAT between **Friday, 2 PM EST** and following **Monday 8PM EST**.
- ✓ UAT Testing starts by following **Tuesday 2 AM EST**.
- ✓ Stories must be in '**Pending release**' Azure board column by **Monday 5PM of the release week** to be included in the release package.

COPADO BEST PRACTICES

Click to add text

COPADO USER STORY BEST PRACTICES

Working with Copado User Stories

- ✓ Avoid User Story dependencies
- ✓ Always develop in designated environment (eg: Dev or Hotfix)
- ✓ DO NOT combine metadata of all/related user stories to a single user story.
- ✓ DO NOT leave empty org credentials in user stories.
- ✓ Avoid manual task requirements, commit code where possible
- ✓ Communicate to DevOps on deployment precedence for user stories, if any.
- ✓ Perform Smoke tests right after SIT Deployment

METADATA COMMITS BEST PRACTICES

Metadata Commits

- ✓ Ensure Dev Org credentials are updated accurately
- ✓ Refresh User story Metadata grid to grab most recent changes in dev box
- ✓ After every commit verify your changes in Azure, only your changes should reflect
- ✓ After selecting Vlocity components, ensure to click '*Get Vlocity Dependencies*'
- ✓ **DO NOT** auto resolve when you promote to SIT. Manually resolve merge conflicts
- ✓ Refrain from making changes to the feature/promotion branches directly in Azure/Git
- ✓ **DO NOT** click refresh button if you are deleting already committed metadata

RE-COMMITS BEST PRACTICES

Metadata Re-commits Sample scenarios

- You have been working on same Apex class throughout the sprint and you have periodically committed your changes.
- There are a lot of changes between the original committed version and the version that is currently working in Dev.

Important

- ✓ Recommit Functionality allows you to recommit previously committed changes.
- ✓ When you select Recommit, previously selected components are selected by default.
- ✓ When you Recommit files, it is highly likely that you may pull in other's changes. Hence be careful if multiple resources are working on same component.

How to Re-commit?

- Bring your User Story to the dev environment by changing Org Credentials manually

RE-COMMIT & RE-CREATE FEATURE BRANCH BEST PRACTICES

Sample scenarios to use Recommit and recreate feature branch functionality

- If you have committed something wrong and want to get rid of the selection, unselect the component, then use Recommit and Re-create feature branch.
- When you have changes in different commits that should be in the same commit due to dependencies (for ex: Profiles and fields): Recommit and Re-create feature branch.

Important:

- ✓ When you Recommit files, it is highly likely that you may pull in other's changes. Hence be careful if multiple resources are working on same component.
- ✓ **Do-not refresh metadata grid** if you just need to remove some previously selected components
- ✓ **When you Re-Create Feature Branch, existing feature branch in the Git repository will be deleted.**
 - ✓ New feature branch will not include the previous commits in the user story. Hence ensure you commit all metadata.

MERGE CONFLICTS BEST PRACTICES

Merge Conflicts

When there is a merge conflict Copado provides 2 options:

1) Copado Conflict Auto-Resolution

- ✓ Copado uses Semantic merge for XML based files. For all other files, A wins over B
- ✓ Git repository keeps documentation of auto-resolved conflicts and can be reverted easily.

2) **Online Conflict Resolution**

- ✓ Copado allows certain metadata types to be excluded from auto resolution.
- ✓ This can be set up either at pipeline or Promotion level
- ✓ This allows developers to review and manually resolve conflict

Smart Conflict Resolution feature

Copado will learn how each conflict have been resolved without prompting you again. The conflict will be marked as AUTO RESOLVED

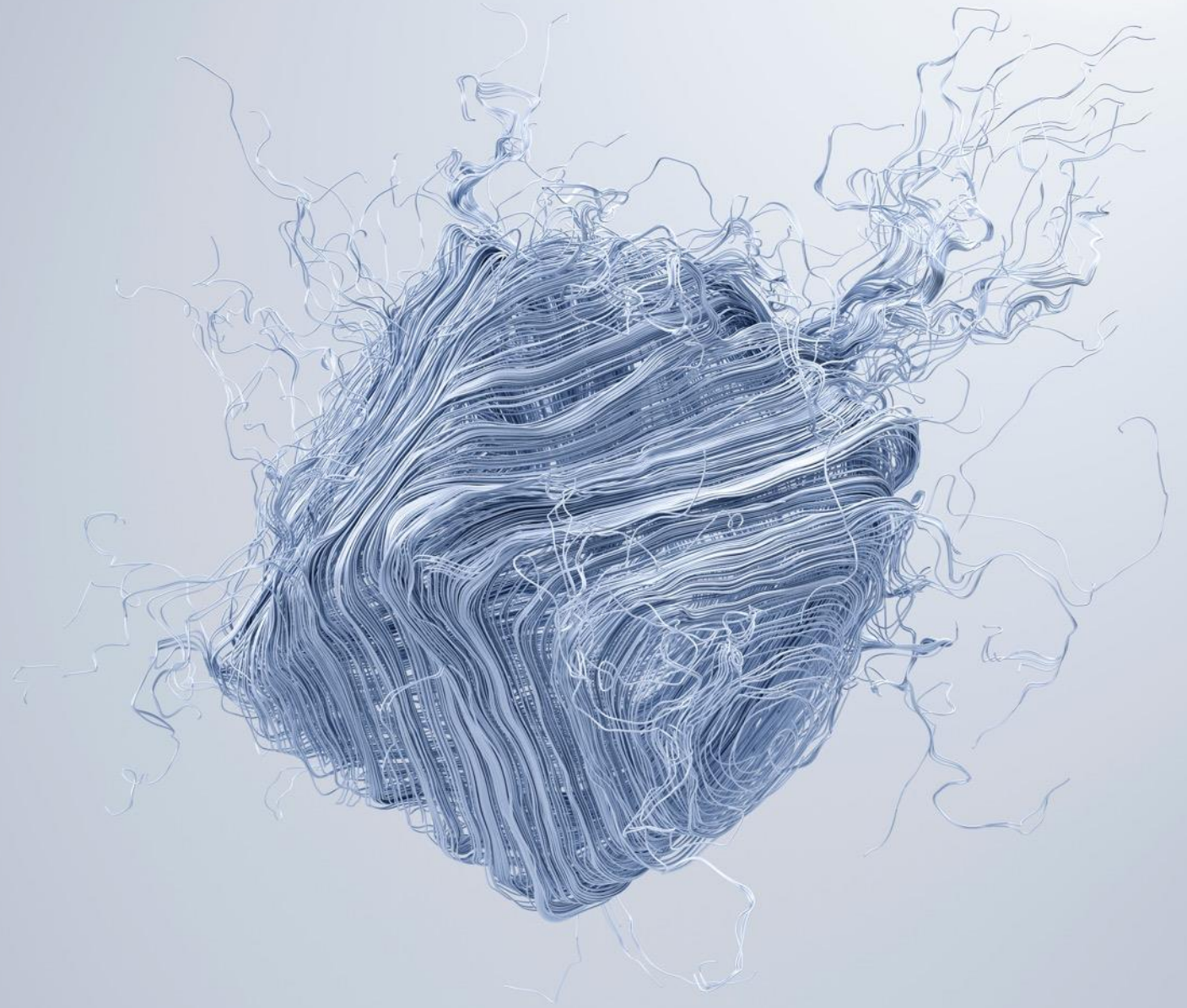
NEXT STEPS



NEXT STEPS

- **REFERENCE** the Developer Best Practice Guide for additional context on Copado How-To's (go/learncopado)
- **REVIEW** quality gates at story level, especially Pull request and SecAPI scans
- **ASK QUESTIONS** as they arise – DevOps Team is here to help!

APPENDIX



What To Master in Source Control

Many post deployment steps can be included in source control because the files these changes reside in are available for import through the metadata API, the interface which allows extraction from SFDC environments to Copado. In reviewing the changes executed in production these past few months, there are opportunities to automate.

Mastered in Source Control

- Applications, App Menus, **Approval processes**, aura, auth providers auto response rules, **assignment rules**
- **Classes**, components, communities, connected apps, custom application components, custom meta data, custom labels, custom permissions, Data category groups, documents
- **Escalation Rules**, email, flexipages, **Flows**, groups, home page components & layouts, **Layouts**, Letterhead
- **Objects – all custom and standard**, object translations
- **Pages, permission sets**, Post Templates, **profiles**
- **Queues**, quick actions, remote site settings, roles
- SAML SSO Configs, sControls, settings, sitedotcoms, sites, static resources, **Triggers**, tabs, test suites, **workflows**, weblinks
- **Vlocity**: AttributeCategory, CalculationProcedure, **DataRaptor**, EntityFilter, IntegrationProcedure, ItemImplementation, **OmniScript**, OrchestrationItemDefinition, OrchestrationPlanDefinition, Pricebook2, Product2, StoryObjectConfiguration, System, VlocityAction, VlocityCard, VlocitySearchWidgetSetup, VlocityUILayout, VlocityUITemplate

Mastered in Production

- Reports
- Dashboards
- Certificates
- Static Resources
- Environment-specific variables (e.g., custom settings, URLs, etc.)
- List Views
- Managed Packages
- Assigning users to Public Groups
- Vlocity Calculation Matrix
- Vlocity Calculation Matrix Version
- [Unsupported Metadata Types](#)